

# US Patent & Trademark Office

## Patent Public Search | Text View

---

United States Patent Application Publication

20250276376

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Nelson; Scott et al.

---

### IN-SITU MODEL COMPARISON FOR ADDITIVE MANUFACTURING SYSTEMS

---

#### Abstract

An additive manufacturing system includes an energy delivery device and a powder delivery device configured to form an as-deposited layer on a build surface of the component. The system includes a topology monitoring system configured to capture data indicative of a position of a surface of the as-deposited layer, and also includes a computing device. The computing device is configured to receive the data and determine an actual position of the surface of the as-deposited. The computing device is configured to compare the actual position to a modeled position of the surface of the as-deposited layer. The computing device is further configured to determine a difference between the actual position and the modeled position of the as-deposited layer and control at least one of the energy delivery device or the powder delivery device based on the difference between the actual position and the modeled position of the as-deposited layer.

---

**Inventors:** Nelson; Scott (Indianapolis, IN), Puhl; David James (Indianapolis, IN), Grafton-Reed; Clive (London, GB), Daum; Peter E. (Indianapolis, IN), Proctor; Robert F. (Indianapolis, IN), Heason; Christopher Paul (London, GB)

**Applicant:** Rolls-Royce Corporation (Indianapolis, IN); Rolls-Royce plc (London, GB)

**Family ID:** 96881616

**Appl. No.:** 18/593417

**Filed:** March 01, 2024

---

#### Publication Classification

**Int. Cl.:** B22F10/85 (20210101); B22F10/366 (20210101); B33Y10/00 (20150101); B33Y30/00 (20150101); B33Y50/02 (20150101)

**U.S. Cl.:**

## Background/Summary

### TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing systems and techniques.

### BACKGROUND

[0002] Additive manufacturing systems generate three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing processes may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing processes may utilize powdered materials, and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures.

### SUMMARY

[0003] The disclosure describes additive manufacturing systems, and methods for operating additive manufacturing systems, that fabricate a component according to deposition parameters. An additive manufacturing system includes an energy delivery device that delivers energy to a build surface of a component to form a melt pool and a powder delivery device that directs a powder stream toward the melt pool to form an as-deposited layer on the build surface. The amount of material captured (e.g., the captured efficiency) or other material or process conditions may vary during an additive manufacturing process, which may lead to the actual dimensions (e.g., thickness) of an as-deposited layer departing from the desired dimensions. Accordingly, the final finished component may depart from specifications. For example, underbuilt and/or overbuilt portions or layers may lead to components that have actual dimension that are less than or greater than the desired dimensions.

[0004] Systems and techniques according to the present disclosure may address these and other problems by inclusion of a topology monitoring system and a computing device. The topology monitoring system may capture data indicative of a position of a surface of the as-deposited layer and send the data to a computing device. The computing device may determine an actual position of the surface of the as-deposited layer based on the received data from the topology monitoring system and compare the actual position of the surface of the as-deposited layer to a modeled position of the surface. The modeled position of the surface of the as-deposited layer may be part of a model stored by the computing device that includes dimensions of each layer of the plurality of layers that make up the component as well as the build strategy (e.g., deposition parameters of the powder delivery device and energy delivery device). The computing device may determine a difference between the actual position and the modeled position of the surface of the as-deposited layer and may control the powder delivery device and/or the energy delivery device based on the determined difference.

[0005] In some examples, the computing device may cause an adjustment to one or more parameters related to the powder delivery device (e.g., a carrier gas flow rate, a powder mass flow rate, or the like) and/or one or more parameters related to the energy delivery device (e.g., a scan speed, a focus, a power supplied to the energy delivery device, or the like). The computing device may determine a relationship between the determined difference and the deposition parameters that create the difference, such as by using machine learning techniques, that may be used to control the additive manufacturing system in the same or subsequent additive manufacturing processes. For example, the computing device may adjust one or more deposition parameters of the powder

delivery device and/or the energy delivery device to reduce the difference between the actual position and the modeled position of the surface of the as-deposited layer. In some examples, the adjustment or adjustments may be made to a toolpath of a deposition head of which the powder delivery device and the energy delivery device are components. In this way, additive manufacturing systems described herein may, by in-situ comparison to a model component, control the powder delivery device and the energy delivery device to produce layers with improved build quality.

[0006] In some examples, the disclosure describes an additive manufacturing system which includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component. The system includes a powder delivery device configured to direct a powder stream toward the melt pool to form an as-deposited layer on the build surface. The system includes a topology monitoring system configured to capture data indicative of a position of a surface of the as-deposited layer, and a computing device. The computing device is configured to receive the captured data from the topology monitoring system, determine an actual position of the surface of the as-deposited layer based on received data from the topology monitoring system, compare the actual position of the surface of the as-deposited layer to a modeled position of the surface of the as-deposited layer, determine a difference between the actual position and the modeled position of the as-deposited layer, and control at least one of the energy delivery device or the powder delivery device based on the difference between the actual position and the modeled position of the as-deposited layer.

[0007] In some examples, the disclosure describes a method for additive manufacturing. The method includes delivering, via an energy delivery device of an additive manufacturing system, energy to a build surface of a component to form a melt pool in the build surface of the component. The technique includes delivering, via a powder delivery device of the additive manufacturing system, a powder stream toward the melt pool to form an as-deposited layer on the build surface. The technique further includes receiving, by a computing device, data indicative of a position of a surface of the as-deposited layer from a topology monitoring system. The technique includes determining, by the computing device, an actual position of the surface of as-deposited layer based on the received data, and comparing, by the computing device, the actual position of the surface of the as-deposited layer to a modeled position of the surface of the as-deposited layer. The technique further includes determining, by the computing device, a difference between the actual position and the modeled position of the as-deposited layer. The technique also includes controlling, by the computing device and based on the determined difference between the actual position and the modeled position of the as-deposited layer, at least one of the powder delivery device or the energy delivery device.

[0008] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

---

## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1A is a conceptual block diagram illustrating an example additive manufacturing system that includes a topology monitoring system.

[0010] FIG. 1B is a conceptual block diagram illustrating example topology monitoring sensors and surface mapping modules of a computing device of an additive manufacturing system.

[0011] FIG. 2 is a process flow diagram illustrating an additive manufacturing technique which includes mass flux and heat flux monitoring and control through comparison to a model of the component technique.

[0012] FIG. 3 is a flowchart illustrating an example method for fabricating a component using

comparison to a model.

[0013] FIG. 4A is a conceptual diagram illustrating an example machine learning model.

[0014] FIG. 4B is a conceptual diagram illustrating an example training process for a machine learning model.

[0015] FIG. 5 is a conceptual and schematic diagram illustrating a portion of an example an additive manufacturing system during an additive manufacturing technique.

[0016] FIGS. 6A-6D are conceptual and schematic diagram illustrating fabrication of an example component during an additive manufacturing technique that employs comparison to a model of the component and control based on the difference between the actual component and the model.

[0017] FIG. 7 is a flowchart illustrating an example method for fabricating a component using comparison to a component model.

#### DETAILED DESCRIPTION

[0018] The disclosure generally describes techniques and systems for monitoring the position of a surface of an as-deposited layer during a blown powder additive manufacturing technique, such as a directed energy deposition (DED) technique, and comparing the determined position of the surface of the as-deposited layer to control fabrication of a component. During blown powder additive manufacturing, a component is built up by adding material in sequential layers, such that the final component is composed of a plurality of layers of material. In blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool, and a powder delivery device may deliver a powder to the melt pool. At least some of the powder at least partially melts and is joined to the melt pool and, thus, the substrate.

[0019] Challenges may arise when performing additive manufacturing techniques with additive manufacturing systems. For example, the amount of material captured by the melt pool, which may be called the capture efficiency, may vary between different layers and or areas of the build surface. For example, the travel speed of a deposition head along a toolpath changing between ends of a track and the center of the track may cause capture efficiency to change, or the complex interaction between the flow rate of carrier gases in the powder stream and coverage gases may change based on the particular position of the deposition head within the system, or the like. The changing amount of material in the powder stream captured by the melt pool may cause variation in the thickness of an as-deposited layer on the build surface, or may cause the thickness of the as-deposited layer to be greater than or less than the desired thickness. The underbuilding or overbuilding of layers may cause the dimensions of the component to fall outside a specification window, causing rework or potential discarding of the fabricated component.

[0020] In accordance with aspects of this disclosure, an additive manufacturing system includes a topology monitoring system. The topology monitoring system is configured to capture data indicative of a position of a surface of the as-deposited layer. The captured data may be delivered to and received by a computing device. The computing device may determine an actual position of the surface of the as-deposited layer based on the received data from the topology monitoring system. The computing device may compare the actual position of the surface of the as-deposited layer to a modeled position of the as-deposited layer and determine a difference between the actual position and the modeled position of the as-deposited layer. The computing device may adaptively control deposition based on the determined difference. For example, the computing device may, based on the determined difference between the actual position and the modeled position of the as-deposited layer, adjust one or more operational settings of the powder delivery device, the energy delivery device, or both. In some examples, the adjustment to the deposition parameters of the powder delivery device and/or the energy delivery device may reduce the determined difference between the actual position of the as-deposited surface and the modeled position of the surface of the as-deposited layer. In this way, systems and techniques according to the present disclosure may compare the actual component being manufactured to a model of the component in-situ (e.g.,

during the additive manufacturing process) to improve quality and/or reduce waste.

[0021] In some examples, substantially no difference between the actual position of the as-deposited surface and the modeled position. In other words, the specification window may be very tight or non-existent, and any determined difference between the actual position and the modeled position of the as-deposited layer may cause the computing device to stop the additive manufacturing process for inspection and/or rework. Alternatively, in some examples, a certain magnitude of difference between the modeled position and the actual position may be acceptable. In such examples, the computing device may determine that the difference exceeds a threshold (e.g., a non-zero threshold) for causing the as-deposited layer to fall outside a specification window. Responsive to determining that the difference exceeds a threshold difference, the computing device may control the energy delivery device and/or the powder delivery device to reduce the distance between the actual position of the as-deposited layer and the modeled position of the as-deposited layer in the same layer or in subsequent layers by adjusting one or more deposition parameters. Thus, the computing device may use the determined difference between the actual position of the as-deposited layer to select or adjust deposition parameters to produce a layer having the desired parameters. In this way, additive manufacturing systems may more accurately produce a layer, and ultimately a component, having the desired parameters compared to systems that do not use in-situ comparison of the actual position of a surface of an as-deposited layer to a modeled position to control deposition parameters.

[0022] The powder delivery device and the energy delivery device may be separate and discrete from each other, or may be combined as parts of a common deposition head. When parts of a common deposition head, the computing device may control the common deposition head to travel along a toolpath across the build surface to deposit a layer. In some examples, the computing device may also individually control the relative position of the powder stream to the melt pool by individually controlling the position of the powder delivery device and/or the energy delivery device within the common deposition head. In some cases, the deposition head may define a central longitudinal axis. The energy may be delivered coincident with or parallel to the central longitudinal axis. In some examples, at least part of the topology monitoring system may include a sensor disposed on the central longitudinal axis. In some examples, at least part of the topology monitoring system may include a sensor displaced from the central longitudinal axis in an “off-axis” position. In some examples, the topology monitoring system may include a combination of on-axis and off-axis sensors.

[0023] The topology monitoring system is configured to capture data indicative of the position of the surface of the as-deposited layer. In some examples, it is considered that the topology monitoring system may consist of a single sensor configured to determine a position of a single point on the as-deposited surface, and project the position of the remainder of the as-deposited surface based on the single point. Determining the position of the as-deposited surface based on a single point may be less computationally intensive than techniques which use multiple points on the surface of the as-deposited layer to determine the position of the as-deposited layer. However, in some examples, the topology monitoring system may include a single or multiple sensors (e.g., image sensors, laser sensors, or the like) configured to determine the position of a plurality of points on the surface of the as-deposited layer. Using multiple points on the surface (e.g., a point cloud) to determine the position of the surface may allow for more accurate and precise determine of the position of the as-deposited surface relative to methods which use only a single point on the surface.

[0024] For instance, the topology monitoring system may be configured to generate a three-dimensional scan of the surface of the as-deposited layer using one or more three-dimensional scanner devices. The one or more three-dimensional scanner device may include at least one of a computed tomography device, a structured-light device, a LIDAR device, or a time-of-flight camera device. Advantageously, generation of a three-dimensional scan of the surface of the as-

deposited layer may allow for more precise selection and tailoring of deposition parameters to match the various points on the surface of the component being fabricated to the model of the component.

[0025] In some examples, the component model includes a computer-aided design (CAD) model that includes the final dimension of the component. The model may also include a position of each as-deposited layer in the plurality of layers that make up the finished component. The modeled position of each layer of the plurality of layers may include a similar three-dimensional scan of each layer (e.g., a modeled point cloud for each layer) for comparison to the captured data representative of the actual position of the as-deposited layer. Alternatively, a single point of along the z-axis (or other build direction axis) may be compared to the actual position to the as-deposited surface.

[0026] The component model may also include other information beyond the dimensions of the component and the dimensions of the individual layers that make up the component. For example, the component model may include a set of deposition parameters for each layer for the energy delivery device and the powder delivery device, which may collectively be called the build strategy. In other words, the model may include the operational settings of the system designed to build each modeled layer in the component model.

[0027] For example, the computing device may set parameters controllable by the powder delivery device that include one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle. Similarly, the computing device may set parameters controllable by the energy delivery device to deposit a layer. The parameters controllable by the energy delivery device may include one or more of a focus of the energy delivery device, a power supplied to the energy delivery device, and the like. In examples where the powder delivery device and the energy delivery device are parts of a common deposition head, the deposition parameters include the travel speed of the deposition head along the toolpath, the amount of overlap between adjacent tracks, and any pauses, dwells, or intermittent movement of the deposition head, or the like. Responsive to determining that a difference in position exists between the modeled position of the as-deposited layer and the actual position of the as-deposited layer, the computing device may adjust one or more parameters controllable by the powder delivery device and/or may adjust one or more parameters controllable by the energy delivery device. The adjustment(s) may reduce a magnitude of the distance between the actual position of the surface of the as-deposited layer and the modeled position of the surface of the as-deposited layer.

[0028] The computing device may compare the actual component being built to the component model and adjust the deposition parameters in an iterative process to maintain control throughout the fabrication process. For example, the topology monitoring system may capture data indicative of the position of the as-deposited surface at the conclusion of the deposition of each layer, and the computing device may select and tailor the build strategy for the subsequent layer to apply the subsequent layer. The as-deposited surface becomes the new build surface, and the computing device may tailor deposition of the subsequent layer to account for any deviation from the model in the previously applied layer. For example, if the captured efficiency of the as-deposited layer was less than expected according to the design, the as-deposited layer may have less thickness than modeled, and the computing device may adjust the deposition parameter to apply the subsequent layer as a thicker layer than originally planned, such that the surface of the subsequent layer will better match the component model because the reduced captured efficiency is accounted for.

[0029] Additionally, or alternatively, the computing device may compare and adjust deposition parameters during deposition of a layer rather than only at conclusion of a layer. For example, the computing device may perform comparison and adjustment based on expiration of a duration of time (e.g., every second, every minute, or the like). In this way, adjustment of deposition parameters may be completed during deposition of a layer, which may improve responsiveness to unexpected changes in the capture efficiency or other process parameters.

[0030] The computing device may control (e.g., adjust parameters) of the powder delivery device and/or the energy delivery device in response to a determined difference between the actual position of the as-deposited layer and the modeled position of the as-deposited layer in one or more ways. In some examples, preset adjustments may be conducted, for example iteratively until difference is eliminated. In some examples, the computing device may store a lookup table which stores both adjustments to settings of the powder energy delivery device and/or the energy delivery device. The computing device may perform an adjustment to the powder delivery device and/or the energy delivery device based on the recommended adjustment from the lookup table. As one example, the computing device may determine that the position of the surface of the as deposited layer is too low, meaning that the as-deposited layer is less thick than the layer is in the component model. Responsive to the determination, the computing device may output a signal that causes the travel speed of deposition head down the toolpath to reduce, or may output a signal that causes an increase of the powder mass flow rate in the powder stream, or output a signal that causes an increase of the amount of overlap between adjacent tracks, or combinations thereof, or the like.

[0031] In one or more examples of the disclosure, the computing device may determine, via a machine learning model that takes the data from the topology monitoring system as input, one or more deposition parameters of the powder delivery device and/or the energy delivery device using one or more machine learning techniques. In this way, the computing device may be trained to execute adjustments to the settings of the powder delivery device and/or the energy delivery device based on previous adjustments that were made and the resulting outcomes of such adjustments.

[0032] Although generally described herein as capturing data indicative of the position of the surface of the as-deposited layer, it is considered that the data captured by the topology monitoring system may be indicative of the position of the build surface. For example, if the as-deposited layer is only partially completed, data indicative of the top surface of the build (e.g., image data) may include data indicative of both the build surface and the surface of the as-deposited layer. In some examples, the computing device may be configured to determine and recognize areas where the position of the build surface is being captured (e.g., those portions on the top surface of the component which have not been visited by the deposition head) and account for the fact that the captured data is of two different layers. For example, an edge of the as-deposited layer may be identified in the captured data, or a temperature difference may be sensed between the surface of the as-deposited layer and the build surface. Recognizing the difference between the two different layers, the computing device may compare the actual position of the build surface to the modeled position of the build surface in areas where data indicative of the build surface is captured, and compare the actual position of the as-deposited layer to the modeled position of the as-deposited layer in areas where data indicative of the as-deposited layer is captured.

[0033] Additive manufacturing systems and techniques of the present disclosure may be used to form any suitable component. For example, the additively-manufactured component may be a component of a gas turbine engine (e.g., a blade, a blisk, etc.), or an additively-manufactured coating on a gas turbine engine component. Forming components or coatings of gas turbine engines in this way may reduce material waste and rework, and/or provide for increased quality relative to other gas turbine engine components formed by other manufacturing processes.

[0034] FIG. 1A is a conceptual block diagram illustrating an example additive manufacturing system **10** configured to monitor a gas flow during an additive manufacturing technique. In the example illustrated in FIG. 1A, additive manufacturing system **10** includes a computing device **12**, a powder delivery device **14**, an energy delivery device **16**, a powder flow monitoring system (PFMS) **18**, an optical system **54**, a stage **20**, a powder source **42**, a powder source mass sensor **44**, and a topology monitoring system **48**.

[0035] Stage **20** is configured to position component **22** during an additive manufacturing process. In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder

delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**. In some examples, component **22** may be considered to be in-situ during any time when component **22** is mechanically supported by stage **20**.

[0036] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream. System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0037] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. **1A**, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. **1A**, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0038] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder and gas flow, such that the combined powder and gas flow defines powder stream **30** focused at a focus plane. The gas flow may include an inert gas, such as argon or nitrogen gas, that maintains an inert atmosphere near melt pool **32**. As powder delivery device **14** is movable in the z-axis shown in FIG. **1A** relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**.

[0039] In some examples, powder delivery device **14** may be mechanically coupled or attached to energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**. In some examples, powder delivery device **14** and energy delivery device **16** may be parts of a common deposition head. Energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.

[0040] In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, and/or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**. In some examples, energy delivery device **16** may be translatable and/or rotatable in three dimensions.

[0041] As shown in FIG. **1A**, energy delivery device **16** may be arranged or configured such that energy **34** and powder stream **30** both exit from a common deposition head and are directed toward



build surface **28**. For instance, energy **34** may pass through a central channel within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder delivery nozzle(s) for forming powder stream **30** and directing powder stream **30** toward build surface **28**. At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**, and at least some of the powder that impacts melt pool **32** may be joined to component **22**.

[0042] During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Energy delivery device **16** may direct energy **34** at first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**, forming as-deposited layer **26**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). In this way the deposition head may traverse across build surface **28** (e.g., back and forth across build surface **28**) depositing tracks of material which form as-deposited layer **26**.

[0043] System **10** may include both mass flow monitoring and heat flow monitoring. To provide mass flow monitoring, system **10** may include powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light sources. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28**. The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz.

[0044] To provide heat flow monitoring, system **10** may include one or more thermal image sensors configured to image at least a portion of melt pool **32** to detect parameters, such as size, temperature, or shape, of melt pool **32**. For example, a thermal sensor may be communicatively coupled to optical system **54** for observing thermal emissions around melt pool **32** and the same or a different thermal camera may monitor a size and/or temperature of melt pool **32**. Optical system **54** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **54** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range. The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device.

[0045] System **10** further includes a topology monitoring system **48**. Topology monitoring system **48** is configured to capture data indicative of a position of surface **36** of as-deposited layer **26**. In some examples, topology monitoring system **48** may include a single sensor which captures data indicative of a position of a single point on surface **36**. Additionally, or alternatively, topology monitoring system **48** may include a single or multiple sensors configured to determine a position of a plurality of points on surface **36** (e.g., as a point cloud).

[0046] By determining the actual position of surface **36** of as-deposited layer **26**, data captured by topology monitoring system **48** may be used to monitor an amount of powder captured by melt pool **32**. For example, system **10** may, by imaging melt pool **32** and the added material, allow the mass to be quantified (e.g., by computing device **12**) using the dimensions of the added material and density of the material (powder). In some examples, topology monitoring system **48** may include a laser and a sensor (e.g., an imaging device), which senses laser light reflected by the

structure being imaged (e.g., melt pool **32** and the added material). The laser may have a defined wavelength, which may affect the resolution of the topology monitoring system **48**. In some examples, the wavelength and sensor may be selected such that the resolution of topology monitoring sensor **48** is as great as about 10 microns (e.g., about 6 microns).

[0047] In some examples, at least one sensor of topology monitoring system **48** may be positioned substantially directly above component **22**, for example on central longitudinal axis L, and may include an interferometer, which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. Additionally, or alternatively, topology monitoring system **48** may include at least one sensor displaced from central longitudinal axis L. As such, topology monitoring system may include at least one sensor may be positioned at an offset with respect to component **22** such that the sensor senses depth information without using an interferometer.

[0048] Topology monitoring system **48** may capture data that allows computing device **12** to generate a three-dimensional scan of surface **36** from the captured data. For example, topology monitoring system **48** may include at least one of a computed tomography (CT) device, a structured-light device, a LIDAR device, or a time-of-flight device. In some examples, topology monitoring system **48** may include a combination of sensors and devices based on one or more of these technologies.

[0049] In some examples, topology sensors of topology monitoring system **48** may be integral with system **10**, e.g., disposed within the enclosure or working area of system **10**. In other examples, topology monitoring system **48** may be an add-on component to system **10**. For example, the enclosure in which the additive manufacturing technique is performed may include a transparent window, and topology sensor **48** may be positioned outside of the enclosure and may image component **22** through the transparent window.

[0050] Topology sensor(s) of topology monitoring system **48** may capture data from a stationary position within system **10**, and may be configured to determine the position of surface **36** based on the distance from surface **36** to the stationary sensor. Additionally, or alternatively, topology monitoring system **48** may include one or more sensors that are configured to move within system **10**. For example, a topology sensor of topology monitoring system **48** may be part of the deposition head that travels along the toolpath to scan energy **34** and direct powder stream **30** to melt pool **32**. In such cases, the mobile sensor may capture data indicative of surface **36**, and computing device **12** may determine the position of surface **36** based on the position of the deposition head and the distance from surface **36** to the sensor.

[0051] Computing device model **12** includes machine learning (ML) model **67**. In some examples, ML model **67** may receive data indicative of surface **36** captured by topology monitoring system **48** as inputs to the machine learning model. In addition to data topology monitoring system **48**, ML model **67** may receive data from optical system **54**, PFMS **18**, and/or other parameters (settings of powder delivery device **14**, settings of energy delivery device **16**, or other sensed data or settings of system **10**).

[0052] Computing device **12** is configured to control components and operation of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** may be communicatively coupled to, and configured to control, powder delivery device **14**, energy delivery device **16**, PFMS **18**, optical system **54**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology monitoring system **48** using respective communication connections. Although FIG. **1A** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**.

[0053] Computing device **12** may be configured to control operation of powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, and/or stage **20**, to position component **22** relative

to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology monitoring system **48** during additive manufacturing processes. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, and/or adjustable z-stage **40** to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology monitoring system **48**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology monitoring system **48** may include positioning a predetermined surface (e.g., build surface **28** to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, and/or topology monitoring system **48**.

[0054] Computing device **12** may be configured to control system **10** to deposit layers **24** and **26** to form component **22** based on a set of deposition parameters. The set of deposition parameters may include energy, feed, and motion parameters that are configured to produce layers **24**, **26**, having various physical parameters, such as a thickness of layers **24**, **26**, (collectively “layers **25**”) and a density of layers **25**. For example, the set of deposition parameters may include power supplied, beam diameter (focus), beam profile, and wavelength of energy delivery device **16**; powder feed rate and gas feed rate of powder delivery device **14**; scan speed and deposition path of stage **20** relative to energy delivery device **16** and powder delivery device **14**; or any other operating parameters that may affect an amount and/or quality of material formed as layers **25**.

[0055] Computing device **12** may be configured to select deposition parameters that are configured to generate layers **25** according to the desired physical parameters. As shown in FIG. **1A**, component **22** may include a base layer **24** and as-deposited layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1A** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including base layer **24** and as-deposited layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers **25**.

[0056] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** according to the set of operating parameters to form, on build surface **28** of base layer **24** of material, an as-deposited layer **26** of material. Computing device **12** may control energy delivery device **16** to deliver energy beam **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**.

[0057] Computing device **12** may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until as-deposited layer **26** is complete. After completion of as-deposited layer **26**, computing device **12** may then control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36**, and this surface will be the build surface for subsequent layers.

[0058] As mentioned above, system **10** may be configured with various in-situ monitoring techniques, including mass flux monitoring, heat flux monitoring, gas flow monitoring, and dimensional monitoring, to control an additive manufacturing process. FIG. **2** is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique. The technique of FIG. **2** may be implemented by system **10** of FIG. **1A** and will be described with concurrent reference to FIG. **1A**.

[0059] Computing device **12** may be configured to control a powder feed rate output by powder

source **42** (see top left of FIG. 2). For instance, computing device **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**. Computing device **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18** and/or powder source mass sensor **44**. Data received from powder source mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**.

[0060] Computing device **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18** and powder source mass sensor **44**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32** into the mass flow determined based on data received from PFMS **18**. Further, computing device **12** may determine an overall mass flux using the data received from PFMS **18** and/or powder source mass sensor **44**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. 2).

[0061] Similarly, computing device may be configured to control energy delivery device **16** to deliver energy beam **34** to base layer **24** to establish a given heat input (see bottom left of FIG. 2). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to base layer **24**, an overlap between adjacent passes of energy **34** across base layer **24**, a pause time between adjacent passes of energy **34** across base layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**).

[0062] Computing device **12** may be configured to receive data from one or more heat sensors, at least one of which may be a melt pool monitor (e.g., a thermal camera), which may be housed within optical system **54**. Computing device **12** may determine a cooling rate and associated heat from using data from optical system **54** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by the melt pool monitor. Computing device **12** may be configured to determine an overall heat flux using these data. Computing device **12** may then use the overall heat flux as an input to the control algorithm used to control the energy delivery by energy delivery device **16** (see top left of FIG. 2).

[0063] Computing device **12** may be configured to the actual position of surface **36** of as-deposited layer **26** during the build, and compare the actual position of surface **36** to a modeled position of surface **36** to validate that system **10** is operating within control. For example, topology monitoring system **48** may capture data indicative of a position of surface **36**, and computing device **12** may determine the actual position of surface **36** based on the captured data. Computing device **12** may also store a model of component **22**. The model of component **22** may include a CAD file including final dimensions of component **22**, and the component model may further store the dimensions (e.g., thickness) of each layer of the plurality of layers **25** that make up component **22**.

Additionally, in some examples, the component model may include the deposition parameters (e.g., operational setting of powder delivery device **14** and/or energy delivery device **16**) that result in the desired dimensions of each layer.

[0064] Computing device **12** may compare the actual dimensions of layer **26** by determining the position of surface **36**, and may compare the actual position of surface **36** to the position of surface **36** stored within the component model. If computing device **12** determines that the actual position of surface **36** matches the modeled position of surface **36** (e.g., does not exceed a threshold difference) the computing device **12** may cause the process to continue based on the current build strategy, that is, according to the deposition parameters stored within the component model.

However, if computing device **12** determines that the difference between the actual position of surface **36** and the modeled position of surface **36**, computing device **12** may adjust control of one or both of powder delivery device **14** and energy delivery device **16**. For instance, computing device **12** may be configured to control a carrier gas flow rate, a powder mass flow rate, and/or a delivery nozzle angle of powder delivery device **14** and/or a focus, a power supplied, a pulse rate, a wavelength, or the like of energy **34** of energy delivery device **16** to reduce the determined difference between the actual and modeled position of surface **36**. In this way, system **10** may include topology monitoring by topology monitoring system **48** to reduce or eliminate deposition of layer **26** at a thickness that falls outside a specification window, or may adjust a thickness of a subsequent layer to bring the build back within the specification window. Thus, topology monitoring system **48** and the associated control techniques may improve build quality of component **22**. In some examples, computing device **12** may control powder delivery device and/or energy delivery device **16** based at least partially on output(s) from ML model **67**, as will be further described below.

[0065] FIG. **1B** is a conceptual block diagram illustrating example topology monitoring sensors which include 3D scanner **49** and interferometer **51** of topology monitoring system **48**. FIG. **1B** also illustrates topology monitoring analysis and control modules of computing device **12** of additive manufacturing system **10**. Computing device **12** may be communicatively coupled to topology monitoring system **48**, optical system **54**, and PFMS **18** to receive data from topology monitoring system **48**, optical system **54**, and PFMS **18**.

[0066] Topology monitoring system **48** includes one or more sensors configured to capture data representative of a portion of surface **36**. In some examples, the one or more sensors may generate sensor data representative of the position of surface **36** in a portion of system **10**, and send the sensor data to computing device **12**. In the example of FIG. **1B**, topology monitoring system **48** includes 3D scanner **49** and interferometer **51**. 3D scanner **49** is a three-dimensional scanner to capture image data that may be used by computing device **12** to generate a three-dimensional scan of surface **36**. 3D scanner **49** may be a computed tomography (CT) device, a structured-light device, a LIDAR device, or a time-of-flight camera device. Interferometer **51** may include a laser and a sensor which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. However, topology monitoring system **48** may include other sensors that are configured to capture data indicative of the position of the same or other portions of surface **36** using the same or other modes. Computing device **12** may be configured to receive data from the one or more sensors of topology monitoring system **48**.

[0067] In addition to data from topology monitoring system **48**, computing device **12** may receive data indicative of various parameters related to powder flow, energy flow, gas flow, and the like of system **10**. For example, PFMS **18** may be configured to detect one or more parameters of system **10** related to powder flow, generate sensor data, and send the sensor data to computing device **12**. Sensor data from PFMS **18** may indicate a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. Similarly, optical system **54** may detect one or more parameters related to energy flow, generate sensor data, and send the sensor data to computing device **12**. Sensor data from optical system **54** may indicate a flow of energy in energy beam **34** between energy delivery device **16** and melt pool **32**.

[0068] Computing device **12** is configured to analyze the sensor data, including the data from topology monitoring system **48**, and control fabrication of a component based on the position of surface **36**. Computing device **12** includes as-deposited surface module **64**, which includes component model **68** and 3D surface module **65**. Computing device **12** also includes an adaptive control module **66**, which includes ML model **67**. FIG. **3** is a flowchart illustrating an example additive manufacturing method for fabricating a component using data from topology monitoring system **48**. Operation of computing device **12** will be described with respect to the method of FIG. **3**. While computing device **12** will be described with respect to modules **64-68**, it will be

understood that various modules may be performed by other computing devices. For example, as-deposited surface module **64** and adaptive control module **66** may be part of separate computing devices, such as specialized computing devices for implementing topology monitoring and/or modelling and machine learning techniques.

[0069] The method includes controlling energy delivery device **16** and powder delivery device **14** (**80**). Adaptive control module **66** is configured to control energy delivery device **16** and powder delivery device **14** according to a set of deposition parameters which, as will be described further below, are at least partly based on data from topology monitoring system **48**. For example, control module **66** may send control signals to energy delivery device **16** and powder delivery device **14** that control energy delivery device **16** and powder delivery device **14** to generate energy **34** and powder stream **30** according to particular parameters. As a result, energy delivery device **16** and powder delivery device **14** may deposit a layer on a build surface based on the control signals (**82**). For example, energy delivery device **16** may deliver energy to a build surface of a component to form a melt pool in the build surface of the component (**84**), and powder delivery device **14** may direct a powder stream toward the melt pool (**86**).

[0070] The method includes capturing data representative of the position of surface **36** of as-deposited layer **26** using topology monitoring system **48** (**88**). For example, topology monitoring system **48** may include 3D scanner **49** that captures image data of surface **36** and/or interferometer **51** that senses data from a laser striking surface **36**. 3D scanner **49** and interferometer **51** may capture data indicative of the position of surface **36** at particular periods to generate a temporal representation of the position of surface **36**, such that the image data may approximate video data or other image data having a temporal component. In some examples, topology monitoring system **48** may capture data at the conclusion of deposition of a layer, prior to deposition of the subsequent layer.

[0071] Computing device **12** may receive data indicative of the position of surface **36** from topology monitoring system **48** (**90**). In some examples, computing device **12** may receive the data as powder delivery device **14** and energy delivery device **16** deliver powder stream **30** and energy beam **34**, respectively, to build surface **28**. For example, the data may reflect conditions and position of surface **36** in real time, such that computing device **12** may use the data as immediate feedback. In some examples, computing device **12** may receive the sensor data after powder delivery device **14** and energy delivery device **16** deliver powder stream **30** and energy beam **34**, respectively, to build surface **28**. For example, image data may reflect the position of the entire surface **36** after deposition of as-deposited layer **26**. In such cases, computing device **12** may use the data as feedback for adjusting operation of powder delivery device **14** and/or energy delivery device **16** to deposit the subsequent layer. In some examples, the method includes receiving additional sensor data, such as from at least one of optical system **54** or PFMS **18**.

[0072] Computing device **12** may determine the actual position of surface **36** based on the received data (**92**). For example, as-deposited surface module **64** may include 3D surface module **65**, which may be configured to generate a three-dimensional scan of surface **36** based on the received data from topology monitoring system **48**. In examples where the received data is a single point, 3D surface module **65** may be configured to project the position of the remainder of surface **36** based on the single point (e.g., set all other points on surface **36** as in the same position in the Z-direction). Alternatively, where the received data is of multiple points on surface **36**, 3D surface module **65** may be configured to stitch together sensed positions of each point in a point cloud to determine the position of surface **36** (e.g., by connecting sensed points according to straight lines or curves).

[0073] Computing device **12** may compare the actual position of surface **36** to a modeled position of surface **36** stored in component model **68** of as-deposited surface module **64** (**94**). Component model **68** may include a CAD file of component **22**, and may further include the desired position of the as-deposited surface at each layer **24**, **26** as the build progresses to form the final version of

component **22**. Component model **68** may include a matrix which includes an ordinal order of layers, and store the corresponding positions of points on each layer. Additionally, component model **68** may include corresponding deposition parameters with each layer that are configured to produce layer **26** which has actual positions on surface **36** that match the modeled positions of surface **36** stored in the matrix.

[0074] Computing device **12** may determine a difference between the actual position of surface **36** and the modeled position of surface **36** stored in component model **68** (**96**). For example, computing device **12** may compare the generated 3D scan of the actual position of surface **36** to a modeled 3D scan of surface **36** stored in component model **68**, and may calculate a distance between the actual and modeled positions at one or more points on surface **36** (e.g., at all points in a point cloud) to determine the difference between the actual and modeled positions.

[0075] Computing device **12** may determine an adjusted set of deposition parameters based on the determined difference (**98**). For example, adaptive control module **66** may be configured to correlate the deposition parameters and/or material parameters necessary to reduce or eliminate the distance between the actual position of surface **36** and the modeled position of surface **36** at each of a plurality of points on surface **36** (e.g., at all points in a point cloud). For example, parameters of the carrier gas, powder flow, energy flow, or deposition head, such as the travel speed of the deposition head along a toolpath, may influence the distance of difference between the actual position and the modeled position of surface **36**. Adaptive control module **66** may adjust some or all of these deposition parameters to selectively tailor the deposition parameters at each point in the point cloud such that the actual position of surface **36** matches the modeled position of surface **36** in component model **68**.

[0076] In some examples, adaptive control module **66** may correlate the determined difference with one or more deposition parameters controllable by powder delivery device **14** and/or energy delivery device **16**. Adaptive control module **66** may be configured to analyze the determined difference between the actual and modeled positions of surface **36** with respect to deposition parameters and determine how a change in deposition parameters affects the difference between the actual and modeled positions of surface **36**. Particular deposition parameters or locations on surface **36** may be associated with an increased difference between the actual and modeled position of surface **36**. For example, a turbulent gas flow at melt pool **32** may cause reduced capture efficiency of powder stream **30** by melt pool **32**, which may cause reduced thickness of as-deposited layer **26**, and thus a difference between the actual position of surface **36** and the modeled position of surface **36**. Adaptive control module **66** may be configured to identify the difference based on the data from topology monitoring system **48** and component model **68**. The data may be used to analyze an effect of changing deposition parameters of powder delivery device **14** and/or energy delivery device **16**.

[0077] In some examples, computing device **12** may be configured to use machine learning techniques to identify differences in the actual and modeled position of surface **36** as correlating to certain deposition parameters, and/or to determine adjustments to deposition parameters that result in a reduction of the determined difference. Such machine learning techniques may enable computing device **12** to adapt and optimize deposition parameters using the observed actual position of surface **36** as indicated by the data from topology monitoring system **48**.

[0078] In some examples, ML model **67** may be configured to correlate the determined difference with the one or more deposition parameters using one or more machine learning techniques. ML model **67** may be configured to determine one or more deposition parameters based on observed behavior indicated by the data using model-based reinforcement learning. ML model **67** may be configured to learn a model of deposition behavior based on data from topology monitoring system **48** and use this model to make decisions about an effect of one or more deposition parameters or change in one or more deposition parameters.

[0079] ML model **67** may identify one or more deposition parameters related to powder delivery

device **14** such as one or more of a carrier gas flow rate, a powder flow rate, or a delivery nozzle position, shape, or other parameter of powder delivery device **14**. Similarly, ML model **67** may identify one or more deposition parameters related to energy delivery device **16** such as a power or focus of energy delivery device **16**. ML model **67** may further receive the data from monitoring system **48** and identify ways that a difference between the actual position and the modeled position may be reduced (e.g., producing a smaller difference distance) or eliminated.

[0080] ML model **67** may include, but is not limited to, reinforcement learning models, such as deep reinforcement learning algorithms or traditional reinforcement learning algorithms. ML model **67** may train the machine learning model using the actual position of surface **36** and the modeled position of surface **36**. The model may learn the underlying patterns and relationships between the deposition parameters and the observed behavior as indicated by the difference between the actual and modeled position of surface **36**. For example, ML model **67** may operate the model to improve a reward signal associated with a reduced distance between the actual and modeled position of surface **36**.

[0081] ML model **67** may determine improved deposition parameters for a given set of conditions. For example, ML model **67** may use optimization methods, such as gradient-based methods or evolutionary algorithms, to tune the deposition parameters. ML model **67** may further validate the model by analyzing the differences between actual position of surface **36** and modeled position of surface **36** of as-deposited layer **26** during fabrication of subsequent layers and/or components.

[0082] ML model **67** may identify one or more parameters that results from or are correlated with a determined difference between an actual and modeled position of surface **36**. ML model **67** may further receive sensor information that provides indication of such parameters, such as temperature data of melt pool **32**, a travel speed of a deposition head along a toolpath, or any other data that may provide an indication of process conditions. As described above, the data input into ML model **67** may include any combination of data from topology monitoring system **48**, parameters determined from the data, other sensed data from system **10**, or various settings of system **10**.

[0083] ML model **67** may determine improved deposition parameters for a given set of conditions. For example, ML model **67** may use optimization methods, such as gradient-based methods or evolutionary algorithms, to tune the deposition parameters. ML model **67** may further validate the model by analyzing determined differences between an actual position and a modeled position of surface **36** of as-deposited layer **26** during fabrication of other layers and/or components, and the resulting build quality that results from reduction or elimination of the difference.

[0084] For example, the deposition parameters may be updated to deposit a reduced amount of material on high areas of a previous layer and deposit a reduced amount of material on low areas of the previous layer. Adaptive control module **66** may determine the adjusted parameters based at least partially on outputs from machine learning model **67**. For example, adaptive control module **66** may generate control signals that cause powder delivery device **14** to modify any of a powder flow rate, a carrier gas flow rate, a delivery nozzle position or angle, or other parameter related to a velocity, directionality, or composition of the flow of powder stream **30** to achieve a powder stream that reduces or eliminates a difference between an actual position of surface **36** and a modeled position of surface **36**. Similarly, adaptive control module **66** may generate control signals that cause energy delivery device **16** to modify any of a power, focus (beam diameter), working distance, pulse rate, dwell time, polarity, or other parameter related to a power, directionality, or wavelength of energy **34** to achieve an energy beam that reduces or eliminates a determined difference between the actual position of surface **36** and modeled position of surface **36**.

[0085] As explained in FIG. **1B** with respect to ML model **67**, additive manufacturing systems described herein may be configured to use machine learning processes for determining a relationship between a determined difference in an actual position and a modeled position of surface **36** and various upstream deposition parameters that cause the determined difference downstream parameters of component **22** that result from the determined difference. FIG. **4A** is a



conceptual diagram illustrating an example machine learning model according to one or more aspects of this disclosure. ML model **67** may be an example of a deep learning model, or deep learning algorithm, trained to determine deposition parameters, material parameters, or component dimensions associated with or influenced by a difference between an actual position and a modeled position of surface **36**. Computing device **12** and/or another device, may train, store, and/or utilize ML model **67**, but other devices of system **10** may apply inputs to ML model **67**. In some examples, other types of machine learning and deep learning models or algorithms may be utilized. For example, a convolutional neural network model of ResNet-18 may be used. Some non-limiting examples of models that may be used for transfer learning include AlexNet, VGGNet, GoogleNet, ResNet50, or DenseNet, etc. Some non-limiting examples of machine learning techniques include Support Vector Machines, K-Nearest Neighbor algorithm, and Multi-layer Perceptron.

[0086] As shown in the example of FIG. **4A**, ML model **67** may include three types of layers. These three types of layers include input layer **102**, hidden layers **104**, and output layer **106**. Output layer **106** includes the output from the transfer function **105** of output layer **106**. Input layer **102** represents each of the input values **X1** through **X4** provided to ML model **67**. In some examples, the input values may include any of the values input into the machine learning model, as described above. For example, the input values may include data from topology monitoring system **48**, various settings and parameters of powder delivery device **14**, various settings and parameters of energy delivery device **16**, various parameters related to component **22**, or another other data received by computing device **12**, as described above.

[0087] Each of the input values for each node in the input layer **102** is provided to each node of a first layer of hidden layers **104**. In the example of FIG. **4A**, hidden layers **104** include two layers, one layer having four nodes and the other layer having three nodes, but fewer or greater number of nodes may be used in other examples. Each input from input layer **102** is multiplied by a weight and then summed at each node of hidden layers **104**. During training of ML model **67**, the weights for each input are adjusted to establish a relationship between data from topology monitoring system **48** and deposition parameters that generate a difference between the actual and modeled positions of surface **36**. In some examples, one hidden layer may be incorporated into ML model **67**, or three or more hidden layers may be incorporated into ML model **67**, where each layer includes the same or different number of nodes.

[0088] The result of each node within hidden layers **104** is applied to the transfer function of output layer **106**. The transfer function may be linear or non-linear, depending on the number of layers within machine learning model **67**. Example non-linear transfer functions may be a sigmoid function or a rectifier function. The output **107** of the transfer function may be a set of deposition parameters, or a set of material parameters, that correspond to a reduced difference between actual and modeled positions of surface **36**. As shown in the example above, by applying ML model **67** to input data such as data from topology monitoring system **48**, processing circuitry of computing device **12** is able to determine a relationship between deposition parameters that produce, and/or component dimensions that result from, the difference between the actual and modeled positions of surface **36** based on the data from topology monitoring system **48**.

[0089] FIG. **4B** is a conceptual diagram illustrating an example training process for a machine learning model according to one or more aspects of this disclosure. Process **110** may be used to train machine learning model **67**. Machine learning model **67** may be implemented using any number of models for supervised and/or reinforcement learning, such as but not limited to, an artificial neural network, a decision tree, naïve Bayes network, support vector machine, or k-nearest neighbor model, CNN, RNN, LSTM, ensemble network, to name only a few examples.

[0090] In some examples, computing device **12** or another device trains machine learning model **67** based on a corpus of training data **112**. Training data **112** may include, for example, image data of previous builds, for examples previous data of differences between actual and modeled positions of surface **36** that did not meet a threshold for causing dimensions of component **22** to fall outside a

specification window, previous deposition parameters or material parameter data, and/or the like. Previous deposition parameter or material parameter data, for example, may include deposition parameters or material parameters associated with previous additive manufacturing processes. In some examples, the data and/or deposition parameter or material parameter data may be generated by executing a natural language processing application on content of a plurality of operational records to automatically extract relevant nor desired training data. For example, by using an NLP model, computing device **12** may capture potentially confounding operation conditions or factors, which may be useful in determining a set of deposition parameters. Computing device **12** may use the NLP model to capture such details. In such an example, the machine learning model may analyze underlying characteristics associated with the powder, rather than powder flow factors, when determining a set of deposition parameters.

[0091] In some examples, training data **112** may include annotations identifying effects of deposition parameters indicated in image data of training data **112**. Training data **112** may include data from past additive deposition processes performed on components having different geometries, formed from different materials, formed under different conditions, and/or the like.

[0092] While training ML model **67**, computing device **12** may compare **119** a prediction or classification with a target output **117**. Computing device **12** may utilize an error signal from the comparison to train (learning/training **118**) ML model **67**. Computing device **12** may generate machine learning model weights or other modifications which computing device **12** may use to modify ML model **67**. For example, computing device **12** may modify the weights of ML model **67** based on the learning/training **118**.

[0093] FIG. **5** is a conceptual and schematic diagram illustrating a portion of an example additive manufacturing system **100** during an additive manufacturing technique. Additive manufacturing system **100** of FIG. **5** may be an example of additive manufacturing system **10** of FIGS. **1A** and **1B**. Similar reference numerals indicate similar elements. Portions of additive manufacturing system **100** are not included in FIG. **5**. These portions will be described with respect to additive manufacturing system **10** of FIG. **1A** and **1B**.

[0094] Additive manufacturing system **100** includes deposition head **156**, of which both energy delivery device **116** and powder delivery device **114** are components. In the illustrated example, energy delivery device **116** includes a laser mounted above or housed in a cavity within deposition head **156**. Energy delivery device **116** delivers energy beam **134** through an opening in deposition head **156** to form melt pool **132** on build surface **128** of previously deposited layer **124** of component **122**.

[0095] Powder delivery device **114** includes channels (not illustrated for clarity) formed within the wall of deposition head **156**. The channels are fluidically connected to delivery nozzles **158A**, **158B** (collectively “delivery nozzles **158**”) from which powder stream **130** is directed to melt pool **132**, adding material to melt pool **132** to form as-deposited layer **126**. As-deposited layer **126** defines surface **136**.

[0096] Additive manufacturing system **100** includes 3D scanner **149**, which is all or a portion of topology monitoring system **148**. 3D scanner **149** is configured to image a portion of surface **136**, and computing device **12** is configured to determine a position of surface **136** based on the captured image. Computing device **12** may compare the determined position of surface **136** to a modeled position of surface **136**, and may control various aspects of system **100** based on the results of the comparison, as described above.

[0097] FIGS. **6A-6D** are conceptual and schematic diagram illustrating fabrication of an example component **22A** during an additive manufacturing technique that employs in-situ comparison to a model of the component and control based on the difference between the actual component and the model. Elements of the actual component **22A** being manufactured are labeled with an “A”. Elements of model component **22M** stored in component model **68** of computing device **12** are labeled with an “M.”

[0098] FIG. 6A illustrates example finished component **222A**, which may be fabricated by system **10** according to techniques of the present disclosure. Final component **222A** includes overall dimensions  $H_o$  and  $W_o$ . Final component **222A** also includes feature **223A**, which includes dimensions  $H_{sub.1}$  and  $W_{sub.1}$ .

[0099] FIG. 6B illustrates model component **222M**, which may be stored as a portion of component model **68** by computing device **12**. Model component **222M** includes layers **225M** configured to make up model component **222M**. Model component **222M** includes overall dimensions  $H_o$  and  $W_o$ . Model component **222M** may also include a position of modeled surface **236M** of layer **226M** making up model component **222M**. Accordingly, model **222M** may define thickness  $H_{sub.2}$  of **226M**.

[0100] FIG. 6C illustrates comparison of actual component **222A** during fabrication with model component **222M** for comparison by computing device **12**. System **10** may deposit as-deposited layer **226A** on build surface **228A**. Computing device **12** may compare the position of surface **226A** to the position of modeled surface **236M**. Computing device **12** may determine a difference  $D_{sub.1}$  between a position of surface **236A** of actual component **222A** and the position of modeled surface **236M**. Although FIG. 6C illustrates comparison and determination of the difference in the Z-direction, the difference may be determined along another axis in cases where the build direction is along that axis. Furthermore, although FIG. 6C illustrates comparison between point A on actual surface **236A** and point M at the corresponding location in the X-Y plane on modeled surface **236M** to determine difference  $D_{sub.1}$ , it is understood that computing device **12** may perform evaluation of the difference at each point of a grid in the X-Y plane, and thus determine differences at each location on the grid.

[0101] FIG. 6D illustrates deposition by system **10** of the layer subsequent to the layer deposited in FIG. 6C. Accordingly, the layer deposited in FIG. 6C retains the cross-hatching pattern as in FIG. 6C, but is renamed base layer **224A** because the subsequent layer is being deposited on build surface **228A** as the as-deposited layer **226A**. Since as-deposited layer **226A** is advanced one level, the surface to which surface **236A** is compared to in component model **68** has advanced one level to surface **237M**. Computing device **12** of system **10** may employ control techniques according to the present disclosure compare the position of surface **236A** to surface **237M** to reduce the determined difference between point A on surface **236A** and point M on surface **237M** to distance  $D_{sub.2}$ . As illustrated, computing device **12** may cause distance  $D_{sub.2}$  to be zero, and may thus eliminate the difference between the actual and modeled positions of the surface. In this way, computing device **12** may bring the build back under control. The additive manufacturing process may continue, depositing layers until final finished component **222A** is produced with dimensions  $H_{sub.O}$ ,  $W_{sub.O}$ ,  $H_{sub.1}$ , and  $W_{sub.1}$  that match (e.g., fall within a specification window) of the dimensions of component model **222M**. In this way, systems and techniques of the present disclosure may reduce material waste and/or may reduce rework by in-situ comparison to a model and adjustment based on determined differences between the actual component being fabricated and the model component.

[0102] FIG. 7 is a flowchart illustrating an example technique according to the present disclosure. The technique of FIG. 7 may be performed by system **10** of FIGS. 1A and 1B, or by system **100** of FIG. 5. The technique may also be performed by other systems, and the described systems may perform other techniques. The technique of FIG. 7 is described with respect to system **10** of FIGS. 1A and 1B.

[0103] Computing device **12** may cause energy delivery device **16** and powder delivery device **14** to deposit layers **25** with prescribed deposition parameters (**302**). After deposition of as-deposited layer **26**, computing device **12** may cause topology monitoring system **48** to capture data indicative of a position of surface **36** (**304**). For example, computing device **12** may cause 3D scanner **49** to capture image data representative of the entire area of surface **36**.

[0104] Next, computing device **12** may determine whether the build is in control (**306**). For

example, computing device 12 generate a three-dimensional scan of surface 36 and may compare the generated scan indicative of the actual position of surface 36 to a position of surface 36 in component model 68 (308). In some examples, computing device 12 may determine whether the difference between the actual of modeled positions of surface 36 exceeds a threshold difference. Responsive to determining that the difference does not exceed a threshold difference, computing device 12 may determine that the build is within control (YES). Computing device 12 may cause system 10 to deposit additional layers of material with prescribed deposition parameters (302). [0105] Responsive to determining that the actual and modeled positions of surface 36 exceeds a threshold difference, computing device 12 may determine that the build is not in control (NO). Computing device 12 may adjust deposition parameters for subsequent layers (310). In some examples, the adjustments may be based on outputs from ML model 67. Computing device 12 may then cause system 10 to deposit an additional layer or layers of material with the adjusted deposition parameters (312). After deposition of the layer or layers, computing device 12 may cause topology monitoring system 48 to capture data indicative of the surface of the additional layers (304). The technique of FIG. 7 may continue until component 22 is completely fabricated. [0106] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0107] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0108] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0109] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0110] Various examples have been described. These and other examples are within the scope of

the following examples and claims.

[0111] Example 1: An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool to form an as-deposited layer on the build surface; a topology monitoring system configured to capture data indicative of a position of a surface of the as-deposited layer; and a computing device configured to: receive the captured data from the topology monitoring system, determine an actual position of the surface of the as-deposited layer based on received data from the topology monitoring system, compare the actual position of the surface of the as-deposited layer to a modeled position of the surface of the as-deposited layer; determine a difference between the actual position and the modeled position of the as-deposited layer; and control at least one of the energy delivery device or the powder delivery device based on the difference between the actual position and the modeled position of the as-deposited layer.

[0112] Example 2: The additive manufacturing system of example 1, wherein, to control at least one of the energy delivery device or the powder delivery device, the computing device is configured to: compare the determined difference between the actual position and the modeled position of surface to a threshold difference; and responsive to determining that the determined difference exceeds the threshold difference, adjust one or more deposition parameters of the energy delivery device or the powder delivery device.

[0113] Example 3: The additive manufacturing system of example 2, wherein, to adjust the one or more deposition parameters, the computing device is configured to adjust the one or more deposition parameters to reduce a distance between the actual position and the modeled position of the as-deposited layer.

[0114] Example 4: The additive manufacturing system of any of examples 1 through 3, wherein the computing device is configured to generate a three-dimensional scan of the surface of the as-deposited layer.

[0115] Example 5: The additive manufacturing system of example 4, wherein the topology monitoring system includes at least one of a computed tomography device, a structured-light device, a LIDAR device, or a time-of-flight camera device.

[0116] Example 6: The additive manufacturing system of any of examples 1 through 5, wherein the powder delivery device and the energy deliver device are parts of a common deposition head.

[0117] Example 7: The additive manufacturing system of any of examples 1 through 6, wherein the computing device is configured to control the powder delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the powder delivery device, wherein the set of deposition parameters controllable by the powder delivery device include one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle.

[0118] Example 8: The additive manufacturing system of example 7, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the powder delivery device; and control, based on the one or more deposition parameters, the powder delivery device.

[0119] Example 9: The additive manufacturing system of example 8, wherein the computing device is configured to determine, based on output of a machine learning model that takes captured data from the topology monitoring system as input, the one or more deposition parameters.

[0120] Example 10: The additive manufacturing system of any of examples 1 through 9, wherein the computing device is configured to control the energy delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the energy delivery device, wherein the set of deposition parameters controllable by the energy delivery device include one or more of a focus of the energy delivery device, a scan speed of the energy delivery device, and a power supplied to the energy delivery device.

[0121] Example 11: The additive manufacturing system of any of examples 8 through 10, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the energy delivery device; and control, based on the one or more deposition parameters, the energy delivery device.

[0122] Example 12: The additive manufacturing system of any of examples 1 through 11, further comprising the component, wherein the component is a gas turbine engine component.

[0123] Example 13: A method for additive manufacturing includes delivering, via an energy delivery device of an additive manufacturing system, energy to a build surface of a component to form a melt pool in the build surface of the component; delivering, via a powder delivery device of the additive manufacturing system, a powder stream toward the melt pool to form an as-deposited layer on the build surface, receiving, by a computing device, data indicative of a position of a surface of the as-deposited layer from a topology monitoring system; determining, by the computing device, an actual position of the surface of as-deposited layer based on the received data; comparing, by the computing device, the actual position of the surface of the as-deposited layer to a modeled position of the surface of the as-deposited layer; determining, by the computing device, a difference between the actual position and the modeled position of the as-deposited layer; and controlling, by the computing device and based on the determined difference between the actual position and the modeled position of the as-deposited layer, at least one of the powder delivery device or the energy delivery device.

[0124] Example 14: The method of example 13, wherein comparing the actual position of the surface of the as-deposited layer to the modeled position of the layer comprises storing a model of the component, wherein the component includes a plurality of layers, and wherein the model includes positions of each as-deposited layer and deposition parameters for each layer of the plurality of layers.

[0125] Example 15: The method of any of examples 13 and 14, wherein controlling at least one of the energy delivery device or the powder delivery device comprises comparing the determined difference between the actual position and the modeled position of surface to a threshold difference; and responsive to determining that the determined difference exceeds the threshold difference, adjusting one or more deposition parameters of the energy delivery device or the powder delivery device.

[0126] Example 16: The method of example 15, wherein adjusting the deposition parameters of the powder delivery device or the energy delivery device reduces a distance between the actual position and the modeled position of the as-deposited layer.

[0127] Example 17: The method of example 16, wherein adjusting the deposition parameters of the powder delivery device or the energy delivery device comprises adjusting the deposition parameters of both the powder delivery device and the energy delivery device.

[0128] Example 18: The method of any of examples 13 through 17, further comprising generating, by the computing device, a three-dimensional scan of the surface of the as-deposited layer.

[0129] Example 19: The method of any of examples 17 and 18, wherein the topology monitoring system includes at least one of a computed tomography device, a structured-light device, a LIDAR device, or a time-of-flight camera device.

[0130] Example 20: The method of any of examples 15 through 19, wherein the computing device is configured to determine, based on output of a machine learning model that takes captured data from the topology monitoring system as input, the one or more deposition parameters.

## Claims

**1.** An additive manufacturing system, comprising: an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool to form an as-

deposited layer on the build surface; a topology monitoring system configured to capture data indicative of a position of a surface of the as-deposited layer; and a computing device configured to: receive the captured data from the topology monitoring system, determine an actual position of the surface of the as-deposited layer based on received data from the topology monitoring system, compare the actual position of the surface of the as-deposited layer to a modeled position of the surface of the as-deposited layer; determine a difference between the actual position and the modeled position of the as-deposited layer; and control at least one of the energy delivery device or the powder delivery device based on the difference between the actual position and the modeled position of the as-deposited layer.

**2.** The additive manufacturing system of claim 1, wherein, to control at least one of the energy delivery device or the powder delivery device, the computing device is configured to: compare the determined difference between the actual position and the modeled position of surface to a threshold difference; and responsive to determining that the determined difference exceeds the threshold difference, adjust one or more deposition parameters of the energy delivery device or the powder delivery device.

**3.** The additive manufacturing system of claim 2, wherein, to adjust the one or more deposition parameters, the computing device is configured to adjust the one or more deposition parameters to reduce a distance between the actual position and the modeled position of the as-deposited layer.

**4.** The additive manufacturing system of claim 1, wherein the computing device is configured to generate a three-dimensional scan of the surface of the as-deposited layer.

**5.** The additive manufacturing system of claim 4, wherein the topology monitoring system includes at least one of a computed tomography device, a structured-light device, a LIDAR device, or a time-of-flight camera device.

**6.** The additive manufacturing system of claim 1, wherein the powder delivery device and the energy deliver device are parts of a common deposition head.

**7.** The additive manufacturing system of claim 1, wherein the computing device is configured to control the powder delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the powder delivery device, wherein the set of deposition parameters controllable by the powder delivery device include one or more of a carrier gas flow rate, a powder mass flow rate, and a delivery nozzle angle.

**8.** The additive manufacturing system of claim 7, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the powder delivery device; and control, based on the one or more deposition parameters, the powder delivery device.

**9.** The additive manufacturing system of claim 8, wherein the computing device is configured to determine, based on output of a machine learning model that takes captured data from the topology monitoring system as input, the one or more deposition parameters.

**10.** The additive manufacturing system of claim 1, wherein the computing device is configured to control the energy delivery device according to a set of deposition parameters that includes one or more deposition parameters controllable by the energy delivery device, wherein the set of deposition parameters controllable by the energy delivery device include one or more of a focus of the energy delivery device, a scan speed of the energy delivery device, and a power supplied to the energy delivery device.

**11.** The additive manufacturing system of claim 8, wherein the computing device is further configured to: determine one or more deposition parameters controllable by the energy delivery device; and control, based on the one or more deposition parameters, the energy delivery device.

**12.** The additive manufacturing system of claim 1, further comprising the component, wherein the component is a gas turbine engine component.

**13.** A method for additive manufacturing, comprising: delivering, via an energy delivery device of an additive manufacturing system, energy to a build surface of a component to form a melt pool in the build surface of the component; delivering, via a powder delivery device of the additive

manufacturing system, a powder stream toward the melt pool to form an as-deposited layer on the build surface, receiving, by a computing device, data indicative of a position of a surface of the as-deposited layer from a topology monitoring system; determining, by the computing device, an actual position of the surface of as-deposited layer based on the received data; comparing, by the computing device, the actual position of the surface of the as-deposited layer to a modeled position of the surface of the as-deposited layer; determining, by the computing device, a difference between the actual position and the modeled position of the as-deposited layer; and controlling, by the computing device and based on the determined difference between the actual position and the modeled position of the as-deposited layer, at least one of the powder delivery device or the energy delivery device.

**14.** The method of claim 13, wherein comparing the actual position of the surface of the as-deposited layer to the modeled position of the layer comprises storing a model of the component, wherein the component includes a plurality of layers, and wherein the model includes positions of each as-deposited layer and deposition parameters for each layer of the plurality of layers.

**15.** The method of claim 13, wherein controlling at least one of the energy delivery device or the powder delivery device comprises comparing the determined difference between the actual position and the modeled position of surface to a threshold difference; and responsive to determining that the determined difference exceeds the threshold difference, adjusting one or more deposition parameters of the energy delivery device or the powder delivery device.

**16.** The method of claim 15, wherein adjusting the deposition parameters of the powder delivery device or the energy delivery device reduces a distance between the actual position and the modeled position of the as-deposited layer.

**17.** The method of claim 16, wherein adjusting the deposition parameters of the powder delivery device or the energy delivery device comprises adjusting the deposition parameters of both the powder delivery device and the energy delivery device.

**18.** The method of claim 13, further comprising generating, by the computing device, a three-dimensional scan of the surface of the as-deposited layer.

**19.** The method of claim 17, wherein the topology monitoring system includes at least one of a computed tomography device, a structured-light device, a LIDAR device, or a time-of-flight camera device.

**20.** The method of claim 15, wherein the computing device is configured to determine, based on output of a machine learning model that takes captured data from the topology monitoring system as input, the one or more deposition parameters.

---